

FAI Governance for Self Architecture Evolution

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Date: May 15, 2026

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Full Aspect Integration (FAI) events produce conflict patterns and action-layer records that can inform governance improvement at every level of a participating Self's architecture. This note formalizes D2.56: how FAI governance interacts with evolution of a Self's integration architecture—the Self-level DNA governing cross-aspect coordination. The note establishes four points. First, Self-level DNA occupies a default home-perimeter position: it does not enter the shared substrate through aspect contribution unless governance makes an explicit decision to include it. Second, two cases permit Self-level content to enter FAI: an explicit governance decision to extend the sharing scope, and a deep strategic partnership with standing configuration and high governance trust. Third—and most importantly—FAI can inform Self-level directed selection even when Self-level content is never shared: conflict patterns at the cross-aspect scope enter the action-feedback locus and feed upward into governance proposals for Self-level integration architecture improvement. Fourth, the Self-level co-adaptation cycle this produces completes the three-tier co-adaptation picture established across D2.40 (instinct layer stays home), D2.55 (aspect-level restructuring), and this note.

1. What this note derives and its position in the series

D2.56 is a cross-paper operational synthesis note. It draws from Paper 3's four-locus evolution-feed mechanism (Claim 4, §7), Paper 3's configuration-as-substrate-content commitment (Claim 5, §8), and Paper 2's three-level governance hierarchy (Self, aspect, cell) with its directed selection, DNA evolution, and action-feedback evolution mechanisms. It extends two prior notes in the D2 sequence: D2.49, which established that Self-level

integration content remains within the home perimeter by default, and D2.55, which formalized the structural co-adaptation cycle for aspect-level restructuring. D2.56 applies the same analytical frame to the level above: the Self-level integration architecture itself.

The question this note addresses is specific: when a Self participates in FAI events over time, how does that participation interact with the evolution of the Self's own cross-aspect coordination rules? The answer has two parts that must be kept analytically distinct. Part one concerns sharing: under what circumstances does Self-level content enter the shared substrate? Part two concerns learning: through what pathway do FAI events inform Self-level integration architecture improvement regardless of whether sharing occurs? Part two is the primary architectural insight; part one specifies the governance conditions under which sharing is permissible.

D2.56 is the third note in an implicit set that maps FAI learning against the three architectural tiers of a participating Self. D2.40 established that the instinct layer is out of scope: LLM weights and instinct-layer content do not cross the inter-Self perimeter, and instinct evolution within each Self continues independently of FAI events. D2.55 established that aspect-level restructuring—reorganization of cells within an aspect or of aspects within the Self—is informed by FAI conflict patterns through the action-feedback and directed selection loci. D2.56 establishes the same for the top tier: Self-level DNA, the cross-aspect coordination architecture.

2. Self-level DNA and its default home-perimeter position

In Paper 2's multi-level governance hierarchy, a Self's DNA layer at the Self scope contains the integration architecture that governs cross-aspect coordination. This is the content that makes the Self a coherent governed entity rather than an uncoordinated collection of aspects. Self-level DNA includes: cross-aspect coordination rules specifying how aspects are integrated into the Self's overall operation; the Self's purpose statement and content-domain specification; expression specifications defining how higher-level governance activates rules within aspects and cells; and vertical integration rules governing how insights and records propagate upward from cells and aspects to the Self level.

By default, this content does not leave the home perimeter. When a Self participates in FAI and contributes aspects to the shared substrate, those aspects carry their constituent cells' DNA-layer and action-layer content into the shared space. The aspect's own DNA content—orchestration patterns, schemas, and rules governing behavior within that aspect's domain—is what becomes available for exchange. Self-level coordination rules are not constituent content of any single aspect; they sit above the aspect level and govern relations among aspects. As D2.49 established, they therefore remain within the home perimeter absent an explicit governance decision to include them.

This default is architecturally appropriate for two reasons. First, Self-level DNA exposes the full integration architecture: sharing it reveals not just one domain of capability but the structural logic that ties all of a Self's capabilities together. This is qualitatively different from sharing an aspect's DNA, which exposes one domain. Second, FAI's exchange-bounding commitment—inherited from Paper 2's instinct/reasoning separation at the inter-Self perimeter—is operative here at the scope of what Self-level content governs rather than at the scope of a single aspect.

The default does not mean Self-level content is inaccessible to FAI events. It means that access requires a governance decision that is distinct from and additional to the governance decision required for aspect-level contribution.

3. When Self-level content may enter FAI

Two cases make Self-level content entry into FAI architecturally permissible.

Case 1: Explicit governance decision to extend sharing scope. Paper 3's configuration-as-substrate-content commitment (Claim 5) includes the sharing scope—which aspects and which levels of content each Self contributes—as a governance-configurable dimension. Governance can configure the sharing scope to include Self-level integration architecture content. This requires an explicit governance decision: the humans holding governance authority over the participating Self must affirmatively choose to extend the sharing scope to the Self level and must confirm, before doing so, their understanding of what Self-level sharing exposes. The confirmation requirement is not procedural; it reflects the architectural fact that Self-level sharing exposes the full integration architecture in a way that aspect-level sharing does not. A governance configuration that includes Self-level content by drift—because no one reviewed what the sharing scope covered—fails to satisfy the governance requirement.

When governance makes this decision, the governance requirements that apply to aspect-level contribution (established in D2.06) apply equally, with the addition of the explicit scope-confirmation requirement. The same provenance carry-over dimensions that apply to aspect-level contribution are configurable at Self level. The same persistence policy dimensions govern what Self-level content remains after FAI dissolution and where.

Case 2: Deep strategic partnership with high governance trust. Two Selves with a standing agreement of sufficient depth and mutual governance trust may include Self-level sharing in their standing FAI configuration. The conditions this requires are those established in D2.29 (high governance trust between participating Selves) and D2.34 (deep cross-organizational agreement), and the standing configuration must be explicit substrate content—not an undocumented operational assumption. A standing configuration that includes Self-level sharing is itself subject to the explicit governance decision requirement in Case 1: the standing configuration must have been established under explicit governance decision, even if subsequent events execute under that standing configuration without re-authorizing Self-level

sharing each time.

Neither case permits Self-level sharing as a default or as a convenience that reduces configuration overhead. The architectural commitment is that Self-level content stays home unless governance deliberately extends the sharing scope.

4. How FAI informs Self-level evolution without Self-level sharing

The primary insight of this note is that FAI events can inform Self-level integration architecture improvement through a pathway that does not require Self-level content to cross the inter-Self perimeter. This pathway runs through the action-feedback locus.

When a Self participates in FAI events, the events produce action-layer records within the Self's home substrate—records of what was attempted, what coordination patterns produced which outcomes, where conflicts arose and how they were handled or preserved. These records are standard action-layer content in Paper 2's sense, and they are processed by the action-feedback evolution mechanism within the home governance authority.

FAI conflict patterns at the cross-aspect scope are a specific class of these records. When FAI events consistently surface conflicts that trace to the way aspects are coordinated rather than to the content of any single aspect, those conflict patterns are evidence about the Self's integration architecture. A Self whose cross-aspect coordination rules produce systematic conflicts with partner Selves' approaches—not because either Self's individual aspect content is problematic, but because their integration architectures interact poorly—will accumulate action-layer records that, if reviewed, reveal this pattern.

The action-feedback evolution mechanism processes these records into proposals. For cell-level and aspect-level issues, proposals target cell DNA or aspect DNA. For cross-aspect coordination issues, proposals can target Self-level DNA—the cross-aspect coordination rules, the expression specifications, the vertical integration rules. These proposals enter the governance decision process as candidates for directed selection at the Self level.

Self-level directed selection operates the same way it does for any other DNA evolution in Paper 2's framework: governance reviews the proposal, evaluates whether the proposed coordination rule change addresses the identified conflict pattern without introducing new problems, and authorizes or declines the change. The authorization is a governance decision over home-perimeter content. No FAI partner's content is being absorbed; the change is to the Self's own integration architecture, informed by patterns observed through FAI participation.

This pathway is architecturally important because it means that the value of FAI for Self-level architecture improvement is available without the governance commitment and exposure that Self-level sharing requires. A Self can improve its integration architecture through disciplined review of FAI conflict patterns, entirely within the home perimeter, with no additional inter-Self governance configuration. Conflict carry-through annotations (established in Paper 3 §6 and

formalized at D1.16) are particularly useful here: when FAI events dissolve with preserved conflicts that span aspect boundaries, those annotations enter home substrate as flags specifically pointing to where the integration architecture may need attention.

Expression mechanism evolution is a specific sub-case worth naming. A Self's expression specifications—how it activates rules in aspects and cells under different coordination contexts—may prove too coarse or too broad when exposed to the variety of coordination contexts FAI events introduce. FAI events that consistently trigger the same expression rules across different contexts, or that reveal that an expression rule activates content that is not task-relevant in the inter-Self context, are informative about expression mechanism design. These observations enter the action-feedback locus and can produce proposals for expression mechanism refinement, which then proceed through directed selection within home governance.

5. The Self-level co-adaptation cycle

The structural co-adaptation cycle at the Self scope follows the same logical form as the aspect-level cycle formalized in D2.55, applied one level up.

FAI events produce action-layer records in the home substrate. Review of those records by governance identifies conflict patterns that trace to cross-aspect coordination rules—the Self-level integration architecture. Governance generates proposals for Self-level directed selection: specific changes to cross-aspect coordination rules, expression specifications, or vertical integration rules that address the identified patterns. Directed selection is authorized or declined under home governance authority. Authorized changes update the Self-level DNA. The updated integration architecture produces better-coordinated aspect contributions in subsequent FAI events, which reduces the frequency and severity of cross-aspect conflict patterns in subsequent action-layer records.

Several structural properties of this cycle are worth making explicit.

The learning is substrate-mediated throughout. FAI conflict patterns enter the action-feedback locus as substrate content; directed selection proposals are substrate content; the governance decision is recorded as substrate content; the updated DNA is substrate content. The cycle does not require inference over undocumented experience; it operates over inspectable records.

The cycle is available at every FAI tier. A Self participating in bilateral FAI with a single partner runs this cycle informed by bilateral conflict patterns. A Self participating in multi-party FAI under joint authority (Paper 3 §4) runs the same cycle informed by a richer conflict pattern set from multiple partner Selves. Population-scale collective evolution (Paper 3 Claim 6) is what this cycle produces when many Selves run it in parallel over accumulated FAI events.

The cycle is asymmetric by design. Each Self's directed selection decisions are made under home governance authority. Two Selves participating in the same FAI events may draw different conclusions about what their respective integration architectures need, may authorize different changes, and may improve at different rates. This asymmetry is not a deficiency; it is the architectural property that preserves each Self's governance autonomy while enabling inter-Self coordination experience to inform internal improvement.

The relationship between D2.40, D2.55, and D2.56 is now complete. D2.40 established that the instinct layer takes no FAI input by architectural commitment—the instinct/LLM boundary holds at the inter-Self perimeter. D2.55 established that aspect-level restructuring is informed through FAI conflict patterns entering the action-feedback and directed selection loci. D2.56 establishes the same for Self-level integration architecture. The three notes together map the full topology of how FAI learning enters the participating Self's governance architecture: it enters at the action-feedback locus for all reasoning-layer tiers (aspect and Self), and it does not enter the instinct layer. The architecture is complete at this description.

6. Anti-pattern: Self-level learning neglect

The anti-pattern this note identifies is conducting FAI events without reviewing whether conflict patterns at the cross-aspect scope should inform Self-level integration architecture improvement.

The failure mode has a characteristic shape. A Self participates in FAI events and processes the resulting action-layer records through aspect-level review. Conflicts that arise within individual aspects are identified and addressed through aspect-level directed selection. Conflicts that arise in cross-aspect coordination are observed but not escalated: they are treated as content problems (addressable at the aspect level), or as partner incompatibilities (requiring negotiation), or as one-off occurrences (not pattern-worthy). Over time, the same cross-aspect coordination patterns produce the same FAI conflict classes, aspect-level adjustments temporarily reduce some symptoms but do not address the coordination logic, and the Self's integration architecture remains mismatched with the coordination contexts its FAI participation requires.

This failure mode is not a result of bad governance decisions. It is a result of the governance review process not looking at the right level. The action-feedback mechanism is operating correctly; the directed selection proposals are being generated at the aspect level; the governance decisions are being made appropriately for aspect-level content. The oversight is that nobody has asked whether the conflict patterns trace to the Self level rather than to any individual aspect.

Closing this oversight requires a governance practice: periodic review of FAI conflict patterns at the Self scope, specifically asking whether patterns that are consistent across multiple aspects and multiple FAI events might indicate integration architecture issues rather than aspect-level

issues. This review is governance work, not automated analysis. The records it requires are already in the home substrate from normal FAI participation. The review does not require additional sharing, additional instrumentation, or additional inter-Self governance configuration. It requires governance attention directed at the right scope.

The anti-pattern is named here because its cost is compounding. A Self whose integration architecture consistently underperforms in FAI contexts accumulates an increasingly large gap between its actual cross-aspect coordination capability and what its FAI participation requires. The gap grows not because the Self is unable to improve but because the improvement mechanism—directed selection at the Self level—is not being engaged. The mechanism is available; the oversight is in not using it.

7. Operational test

For a Self with an established FAI participation record, the following test determines whether Self-level co-adaptation is being practiced.

A qualified observer examines the Self's governance substrate for Self-level directed selection events—recorded governance decisions that modify cross-aspect coordination rules, expression specifications, or vertical integration rules. For each such event, the observer examines the reasoning recorded in the governance substrate. The observer asks: does the reasoning reference FAI conflict pattern analysis? Specifically, does it identify a cross-aspect coordination issue, trace that issue to a pattern across multiple FAI events, and cite the pattern as the motivation for the Self-level directed selection?

If Self-level directed selection events exist and their governance reasoning references FAI conflict pattern analysis, the Self is running the co-adaptation cycle described in §5. If Self-level directed selection events exist but their reasoning does not reference FAI conflict patterns—if they are driven entirely by within-Self analysis, strategic priority changes, or other sources—the observer cannot conclude that the co-adaptation cycle is running; the learning connection may be absent. If Self-level directed selection events do not exist despite sustained FAI participation, the anti-pattern described in §6 is a candidate explanation.

The test requires that governance reasoning be recorded in the substrate—a requirement that follows directly from Paper 2's directed selection mechanism and Paper 1's path retraceability commitment. A Self whose governance decisions are not recorded as substrate content does not satisfy the Paper 1 and Paper 2 commitments that this note derives from and therefore does not satisfy the operational conditions this test assumes.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *FAI Governance for Self Architecture Evolution*. May 15, 2026. ORCID: 0009-0004-8065-3235.